Python Programming - 2301CS404

Lab - 4

Roll No.:324

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01) WAP to print 1 to 10.

In [2]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

02) WAP to print 1 to n.

In [6]:
```
num = int(input("Enter Number:"))
for i in range(1,(num+1)):
    print(i)
```

```
Enter Number:45
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
```

03) WAP to print odd numbers between 1 to n.

In [7]:
```
num = int(input("Enter Number:"))
for i in range(1,(num+1)):
    if i%2!=0:
        print(i)
```

```
Enter Number:10
1
3
5
7
9
```

04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

In [8]:
```
num1 = int(input("Enter Number:"))
num2 = int(input("Enter Number:"))
for i in range(num1,(num2+1)):
    if i%2==0 and i%3!=0:
        print(i)
```

```
Enter Number:10
Enter Number:15
10
14
```

05) WAP to print sum of 1 to n numbers.

In [9]:
```
num = int(input("Enter Number:"))
sum = 0
for i in range(1,(num+1)):
    sum+=i
print(sum)
```

```
Enter Number:10
55
```

06) WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

In [11]:
```
num = int(input("Enter Number:"))
sum = 0
for i in range(1,(num+1)):
    sum+=i*i
print(sum)
```

```
Enter Number:4
30
```

07) WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

In [13]:
```
num = int(input("Enter Number:"))
sum = 0
for i in range(1,(num+1)):
    if i%2==0:
        sum-=i
    else:
        sum+=i
print(sum)
```

```
Enter Number:2
-1
```

08) WAP to print Multiplication Table of the given number.

In [14]:
```
num = int(input("Enter Number:"))

for i in range(1,11):
    print(f"{num} * {i} = {num*i}")
```

```
Enter Number:5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

09) WAP to find Factorial of the given number.

In [16]:
```
num = int(input("Enter Number:"))
ans=1
for i in range(1,num+1):
    ans*=i
print(ans)
```

```
Enter Number:4
24
```

10) WAP to print GCD of given two numbers.

In [38]:
```
num1 = int(input("Enter Number:"))
num2 = int(input("Enter Number:"))

value = min(num1,num2)

ans = 0

for i in range(1,value+1):
    if num1%i==0 and num2%i==0:
        ans = i
print(ans)
```

```
Enter Number:10
Enter Number:15
5
```

11) WAP to find Factors of the given number.

In [17]:
```
num = int(input("Enter Number:"))

for i in range(1,num+1):
    if num%i==0:
        print(i)
```

```
Enter Number:10
1
2
5
10
```

12) WAP to find whether the given number is Prime or not.

In [20]:
```
num = int(input("Enter Number:"))

for i in range(2,num):
    if num%i==0:
        print(f"Not Prime")
        break;
else:
    print(f"Prime Number.")
```

```
Enter Number:7
Prime Number.
```

13) WAP to print sum of digits of given number.

In [23]:
```
num = int(input("Enter Number:"))
sum=0
while num!=0:
    sum+=num%10
    num//=10

print(sum)
```

```
Enter Number:123
6
```

14) WAP to check whether the given number is Palindrome or not.

In [25]:
```
num = int(input("Enter Number:"))
temp = num
ans = 0;
while num!=0:
    ans = ans*10+num%10
    num//=10
if ans == temp:
    print("Palindrome")
else:
    print("Not Palindrome")
```

```
Enter Number:123
Not Palindrome
```

15) WAP to check whether the given number is an Armstrong Number or not.

In [31]:
```
num = int(input("Enter Number:"))
count=0
temp = num
```

```
sum=0
while num!=0:
    count+=1
    num//=10
num=temp
while temp!=0:
    sum+=((temp%10)**count)
    temp//=10
if sum==num:
    print("Armstrong.")
else:
    print("Not Armstrong.")
```

```
Enter Number:370
Armstrong.
```

16) WAP to print all the perfect numbers between 1 to n.

In [3]:
```
num = int(input("Enter Number:"))



for i in range(1,num+1):
    sum=0
    for j in range(1,i):
        if i%j==0:
            sum+=j
    if sum==i:
        print(i)
```

```
Enter Number:100
6
28
```

In [ ]: